taken care of. These Act and Ordinance are designed by the Ministry's National Industrial Safety Council of Nigeria (NISC) to promote industrial safety and prevent hazards. NISC also disseminate information on how to avoid accidents particularly those involving employees.

The factory Act is outlined into these groups viz. Health and Safety; for health provisions including among others as follow: Cleanliness, Overcrowding, Temperature, Ventilation, Lighting, Drainage, Sanitary, Accommodation while for safety provisions include among others as follows: Fencing (guards), Cleaning, Training, Hoist, Lift, Chains, Ropes, Lifting tackle, Cranes, Floors, Access, Equipment, Fire, Eye protection, Protective equipment. Another outline of the Act is Welfare provisions which include among others as follows: Drinking water, Washing, Clothing, Accommodation, Seating facilities, First aid box and so on.

(a) **Cleanliness:** This is also known as Housekeeping. The employer must see that the factory environment are kept clean always to check the spread of disease. Floors and benches, and so on, must be clean daily, washed and dried.

b) **Overcrowding:** There must be enough space i.e. at least $40m^2$ of space for every person and not less that 4m high. Each machine must be well spaced from one another and enough walking and working space provided.

(c) **Temperature:** This must be reasonable.

(d) **Ventilation:** All work areas must be adequately ventilated in order to provide fresh air. Worker must be protected against dust, fumes and other impurities. Ventilation could either be secured by natural or artificial means.

(e) **Lighting:** Must be suitable in every part of the factory where people are working. Lighting could either be natural or artificial. It should not be obstructed unless otherwise instructed. Areas where employees pass in course of their employment should also have lighting. Windows and skylights must be kept clean always.

(f) **Drainage:** Must be provided where wet processes are carried out and during rainy seasons or rainfall.

(g) **Sanitary Accommodation:** Must be provided for each sex separately and be kept clean and lighted. This provision must be suitable, sufficient and convenient.

(h) **Fencing:** Dangerous rotating parts of machines, engines, and so on, must be effectively fenced (guards) unless so constructed to be in a safe position. The guards must be properly maintained always. Rotating parts that require fencing are belts, gears, sprockets, chains, flywheels, electric generators/motors, flanges, pulleys.

(i) **Cleaning:** Factory, workshops, and so on, floor must be cleaned; wipe off lubricating oil or other liquids on the floor; removal of obstructions on the floor; machines, engines, and so on, to be cleaned. Stocks to be properly piled up to avoid obstruction; gangways and aisles must be cleaned and clear off obstacles.

(j) **Training:** New employees must be trained and supervised for any work done on any machine that is likely to be dangerous.

(k) **Hoists and Lifts:** Must be sound, properly maintained and examined frequently. Must be suitably enclosed and their safe-load weights must be adhere to.

l) **Chains, Ropes and Lifting Tackles:** Must be soundly constructed and examined frequently. Their safe lifting loads must be adhere to.

(m) **Cranes:** Must be soundly constructed and properly maintained. Their safe working loads must be adhere to.

(n) **Floors:** These includes steps, stairs, gangways, aisles and buildings must be properly and soundly constructed. They must be maintained regularly. Handrails must be provided for stairs.

Access: There must be a safe means of access (Passages) to and fro all work places.

3

(p) **Equipment :** These includes plants, machines, engines, and so on, which must be sound, properly installed and maintained. They should be made to run according to their specifications.

(q) **Fire:** Every factory must have alarms and fire-fighting equipment. It is the duty of the employer to provide adequate means of escape in case of fire, which must be kept free from obstructions and marked conspicuously. A factory have a certificate from the Fire Authority (FA) stating that, the means of escape are reasonable. Fire is caused by three agents in combustion combinations viz. Heat i.e. from external source or chemical reactions; Fuel i.e. either solid, liquid or gas and Oxygen i.e. from surrounding air which contents about 21% oxygen. If one of these agents is taken away, there will be no fire, hence the production of fire-extinguishers. Fire-extinguishers helps to cool the burning materials below its ignition point; to smoothers the fire by excluding oxygen and to remove the extra fuel. Fire is divided into three classes, so also is the fire-extinguishers.

(i) Class A Fire

This is an ordinary fire caused by burning of wood, paper, grass, textile and so on, and can be put-off by class A fire-extinguisher. This class of extinguisher uses water as a quenching substance. In such a fire, in absent of water, soda acid can be used. Never should one try to quench electric or flammable liquid fire with water or soda acid fire-extinguishers. It could result to electrical shock and more spread of the fire.

(ii) **Class B Fire**

This is that fire involving flammable liquids such as oil, petrol, paints, alcohol, spirit, gasses, and so on, and can be easily quenched by using those fire-extinguishers using quenching substances like dry-power, carbon dioxide, foam, vaporising liquids.

(iii) Class C Fire

This is that fire involving electrical apparatus, and so on, which is also called "electrical fire". Such fire can be easily quenched by using those fire-extinguishers

V

using such quenching substances like dry powder, carbon dioxide and vaporising liquids. Before one fight electrical fire, the power supply must be put-off first. These quenching substances are non-conductive agents which must be used always.

### (iv) Operation of Fire Extinguishers

Try to understand the operating instructions written on the body of the extinguisher placed at a conspicuous area in the workshop or factory floor. Learn how to operate it, else you may not have the time to read the instructions when face with an outbreak of fire. Ensure that access to all extinguishers are not obstructed.

**Water or Soda Acid Extinguisher:** Remove the cover cap, turn extinguisher upside down, strike the knob or plunger and liquid is forced out under pressure, directing the hose to source of fire.

**Carbon dioxide:** Pull out the safety pin and press the trigger. In doing so, a vigorous stream of gas is emitted; directing hose to source of fire.

**Dry powder:** Same as for carbon dioxide.

**Foam:** Turn the handle and tip upside down or remove the cap and strike the plunger; directing the hose to source of fire.

**Vaporising Liquids:** It smoothers the fire. Either pump the handle or strike the knob. In doing so, a heavy vapour is released and direct the hose to source of fire. It is very common among vehicle fire-extinguishers.

### (r) Protective equipment

The need for protective wear is obvious when one consider the dangerous work situations inherent in utilities and workshop duties. Employers must provide protective wear for their employees and enforce it on them to use these wears in the various floors and sections in their set up. To protect the head, the suitable safety item is the hard hat otherwise called safety cap. There are two types of these hats viz. full brim; which is completely around the head, extending over the forehead, ears and back of the neck and the other one is the short brim; visor; with a peak in the front. The later is commonly used in factories because it is more compact and easy to store. Also, it is less apt to tilt or catch and not easy to fall from the head in high winds for it is held firmly in position by chin strap.

To protect the foot, one need to wear a safety boot. The best one is the boot that have a steel toe-box which is capable of supporting less than 50kg of static load. A blow on the arch of the foot, the shoe can be able to withstand it.

To protect the body, one need to wear an over-all otherwise called working jackets. The common ones are the jacket type and the cover-all type.

To protect the eyes, one need to put on the goggles when doing such works like grinding, chipping, masonry. Goggles in general use, are the cover-all, the visor antiglare and the spectacle types. The cover-all goggle is commonly used in most workshops and factories. [redacted]

(s) **Drinking water:** Must be supplied in wholesomely and adequately.

(t) **Washing:** Bathrooms must be provided. Clean running water (hot or cold) must be provided. Soap washing and towels for cleaning hands respectively must be provided and properly maintained.

(u) **Clothing accommodation:** Cloak-rooms must be provided in order to accommodate clothes which are not worn during working hours. It should be adequately lighted and ample space provided.

(v) **Seating facilities:** When there are reasonable opportunities for sitting down during working hours without detriment to the work, suitable seats must be provided.

(w) **First aid Box:** There must be a first aid box or cupboard mounted on the factory floor placed in charge of a responsible person who has been trained in first aid. This takes care of any casualty before being placed under medical care.

## 1.3 HINTS AND PRECAUTIONS ON MAINTENANCE HAZARDS

Most accidents that normally occur in our workshops or factory can be divided into three major groups. They are Mechanical hazard; Electrical hazard and Non-Mechanical hazard. Generally, objects left lying around the workshop floor can cause people to trip and fall. Spilled oil on the floor can cause people to trip and fall. Spilled toxic vapour emitted by people can become a risk to health. Combustible waste, dust accumulator can cause a explosion and create fire hazard. Handling materials stored in a dirty environment can lead to skin infections. Cluttered work space creating a cramped environment can lead to injury and damage to equipment.

**Mechanical hazard:** This is the accident associated with the use of machines. Avoid wearing loose clothes or ornaments that can entangled to machines rotating parts. Never walk away from running machines. Isolate machines at the mains when not in use. Ensure to remove chuck-keys, wrenches and any other metals from the machines before it is put on operation. Do not operate a machine without the fence (guard) and always re-place guard when want to start machines. Remove swarfs with brush and always adopt correct procedures.

**Electrical hazard:** Maintenance crew could receive electric shock, direct burns and injury through electricity. To avoid all these hazards, one must ensure to use a square pin fused main plug, not to overload sockets and beware of naked electrical wires. Equipment must be isolated from mains, preferrably, remove the fuses whilst maintenance takes place. Never use power tools in wet and damp condition. Ensure to examine the lead and plug for loose wire connections or other damages in power tools. Ensure that external metal parts of equipment should be permanently connected to earth.

Non-Mechanical hazard:

(a) **Chemical hazard:** Always label bottles cylinders and containers clearly marked emphasising the type of chemical. Do not eat or smoke in this area. Always wash your hands when finishing work. Never pipette chemicals by mouth. Ensure to wear safety items. Do not store incompatible chemicals in close proximity to each other.

(b) **Fire hazards:** Store flammable liquids from main factory or workshop and marked "highly inflammable". These liquids should not be kept indirect sunshine/light. Provide the correct type of fire-extinguishers and apparatus. Combustible waste must not be allowed to accumulate. Stair-ways, gang-ways, doors and other exit should not be obstructed. Oily and grease clothes should be kept in a close-up dust bin.

(c) **Manual hazard:** These are accidents caused when lifting and carrying objects manually in the workshop or factory floors. Always position the feet correctly to ensure good balance. Crouch as close to the object. Grip the object firmly. Always keep the back straight and straighten the legs when lifting. Use crowbars to raise heavy loads for machine lifting. Use rollers to transport heavy loads if there are no cranes, and so on, for lifting purposes. Do not obstruct your view or vision when carrying an objects.

(d) **Tool hazard:** Never must you use over-size spanners of wrenches on bolt heads and nuts. Usecorrect tools always viz. a file with its handle; not to use a mushroom head chisel; a sound and better fitted hammer head handle.

(e) **Climbing hazard:** Ensure to use sound and well constructed ladder. Always use the safety belt and ensure that the ladder is well supported before climbing.

## 1.4 CAUSES OF ACCIDENTS

Accidents could be avoided but due to the followings, most accidents are recorded in the workshop and factory floors.

(a) **Carelessness:** Accidents are caused by carelessness of the operator, work-mates and even the employer. Concentration on work being done eliminates the possibility of most accidents.

(b) **Ignorance:** Lack of proper knowledge of the machine used by the employees result into accident. Hence this safety regulation of "if you do not know, ask" must be adhere to.

(c) **Unsuitable clothing:** Loose sleeves, unbuttoned or-torn clothes and flapping ties result in accident and must be avoided.

(d) **Untidiness:** Poor housekeeping can result into accident.

(e) **Over-confidence:** Too know on the part of employee can result into accident. Always adhere to the proper ways in operating a machine of carrying out duties. There is nothing like bad luck or home trouble. Guide against over-confidence.

(f) **Tiredness:** Never should an employee operate a machine when he is tired or to carry out any other duties. Operating a machine or doing any other duties when energy is exhausted can result into accident.

## 1.5 COST OF AN ACCIDENT

The cost of an accident can be referred to that cost to an employee, and that cost to an employer which are in terms of total cost in money, material, physical and other considerations.

(a) **Employee:** Loss of earning could occur if the accident that had happened is link to him. Loss of confidence and morale could occur if the employee was injured in the accident. This could make him feel no more competent on the job again and probably to be feared of his future job. Physical disability could occur if the employee who was injured in the accident losses any part of his body. This could result to the employee not able to work normally again and probably losses his job. Death could occur if the employee who was injured in the accident and later died. This is an ultimate cost to the employee as he lost his life.

(b) **Employer:** Lost of wages could occur if the employer is responsible for the accident. Employer is by law will pay wages to the employee. Plant cost could occur if the plant involved in the accident is damaged. This will facilitate investigation into the cause of the accident and bring about repairs to the damaged plant or replacing the wrecked plant. As a result of an accident and the strength of damage and injury, the insurance premium may be increased by the insurer in order to guide against future cost of repairs and compensation.

## 1.6 SUMMARY 30/03/23

It is necessary for any employee to obey and observe all safety regulations of the factory or workshop they find themselves in. Report all known accidents to the officer in-charge of the floor or section either minor or fatal. In doing so, one is helping to find a solution to remove a repeating accidents of such a nature. As for the employers it is their duties to provide and ensure (force) that safety precautions are abide to. Draw up a time-table to teach employees the need of safety awareness in the factory or workshop. It is essential for Factory Inspectors to visit each factory or workshop to see if safety equipment are supplied to employees installed in the factory or workshop premises and if employees are using them. Other protective equipment include the ear safety device to prevent noise from an operator's ear drums, being damaged. It is also necessary to wear respirator if one is working in a chemical or dusty floor or section so that pure oxygen can be taken in. Hand gloves must be worn by employees working in areas that the hand palm or fingers could be injured.

Protective appliances should be provided and must be used by employees in any processes which will expose them to danger. Devices to regulate the functions of the machines must be provided in order to cut-off power in case of emergency. Employees must guide against environmental hazards especially objects left lying around and combustible waste. Employees must ensure that the over-alls they are wearing are cleaned regularly and free from oily or greasely nature, as such conditions could attract fire hazard. Employers should ensure that reviews or investigations carried out in time of accidents either by the employers or safety inspectors should be minimised because they take up both supervision and administration cost. Employees should also remember that there is no work so important, so great, so rewarding either in normal duties or emergency-period that SAFETY should not be recognised.

In establishing a safe environment, one would avoid injuries and loss of life. Employer that is safety conscious contribute to staff morale, which improve the standard and rate of work done. It reduces damage to equipment and properties and burden on health services. Accident free environment encourages more profit margin, would be employees to apply for services and loss of production and salary are avoided.

## SELF CHECK

1. Write short notes on the followings:

   (a) Factory Act (b) Safety Ordinance

2. List five protective equipment and sketch three from those listed.
3. What is Fencing in Factory Act?
4. List three safety regulations of manual handling with reference to safety ordinance.
5. Write short notes on the following using sketches:

   (a) Goggles (b) Safety cap (c) Safety boot and (d) Over-all.

6. State four causes of tool hazards
7. State the operation of vaporising liquid fire-extinguisher in fire fighting.
8. Write short notes on the following causes of accidents:

   (a) Ignorance (b) Over-confidence (c) Tiredness

9. What is Washing in welfare provision of the Factory Act?

## CUTTING FLUID

A Metal Machining is accompanied by excessive (heat) and friction between tool and workpiece which thus produces great heat. This lowers the productivity of the process by wearing out the tool and producing poor work surfaces and dimensional accuracy.

These effects are eliminated or minimized by the application of cutting fluid.

✓ A cutting fluid is any fluid which cools and lubricates the tool and workpiece in a machining operation. Therefore a cutting fluid can improve the efficiency of a machining operations by the following means.

(i) Cooling the cutting tool and the workpiece
(ii) Lubricating the cutting and non cutting surfaces of the tool
(iii) Inhibiting seizure between the chip and tool.
(iv) Flushing out the chips from the work area.

### BENEFITS OF USING CUTTING FLUIDS.

(i) Brings about improved surface finish
(ii) Better control of workpiece accuracy.
(iii) Lower tool force.
(iv) Rust prevention on the newly machined surface.
(v) Faster cutting speed or improve tool life.

✓ Most practical cutting fluids have vegetable or mineral oils base, although mineral oil is the most widely used. Some of these oils are used as straight oils, while the rest as emulsion with water. Oil-in-water (o/w) emulsion finds application where cooling is the most important consideration, because they have such heat conducting capacity than straight oils. This emulsion is used with almost 90% of machine tools. Straight oils are used when lubrication is most important consideration.

When the general purpose cutting fluid is not readily available the following formulation derived locally may be used [30]: 9% of ordinary domestic washing soap, 9% emulsifying wax, 0.075% potassium dichromate, 0.075% of naphthol, and 0.05% sulphur. All mixed together and dissolved in 90% low grade automotive oil at 60°C temperature.

## DRILLING MACHINE.

The drilling machine is used for producing holes in a workpiece. It is a very important machine tool which is commonly used in the workshop. The holes made may be through or blind. The common tool used on the machine is the twist drill. The holes produced may finished or modified by boring, reaming, counter sinking, counter boring, tapping or spot facing.

The drilling machine is made up of essentially the base, column, spindle, table, feed mechanism and the adjustable head or sleeve.

(1) BASE. This carries the whole machine structure. It is a casting with slots for bolting the machine to the floor or to the pedestal.

(2) COLUMN: This supports the drill press head and also the table on some machines. The column are usually circular in section.

(3). SPINDLE: This carries the drill or other cutting tools and revolves in a fixed position in the sleeve.

(4) TABLE: This is supported on an arm which is adjustable on the lower section of the column. The table is adjustable vertically

to accomodate different heights of work. It May also be Swinged out of the Way So that the work Can be Seated on the base if required.

## TYPES OF DRILLING MACHINES. ✓

The common types of Drilling Machines includes the following:

1. Manual feed Vertical Drilling Machine
2. Power feed Vertical Drilling Machine
3. Radial Drilling Machine
4. Multi-Spindle Drilling Machine.
5. Portable drilling Machine.

## DRILLING MACHINE TOOLS. ✓

These tools includes:

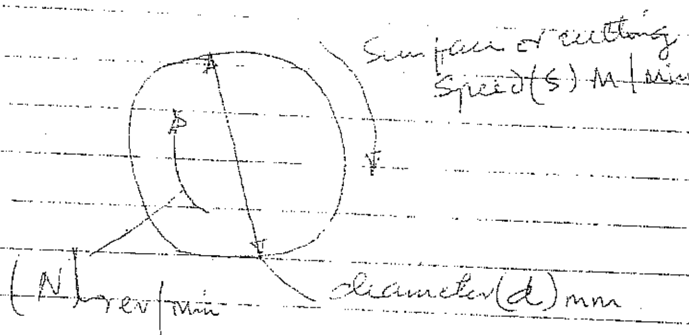
Twist drill, Counter Sinks, Counterbores, Spot facers, Core Drills and reamers.

## CUTTING SPEED. ✓

The relative speed between the cutting tool and work piece is known as the cutting speed and is expressed in Meters per Minute.

As an example, we shall find the cutting speed (S) of a workpiece of diameter (d) mm being turned at (N) rev/mm as can be seen below:

The distance travelled by a point on the circumference = $\pi D$ mm.






911

Consequently in (1) one Minute, the work piece turns through (N) revs, So that the distance travelled by the Same point in one Minute = $\pi dN$ mm/min

But cutting Speed (S) is expressed in Meters per Minute, So dividing by 1000, we get

cutting Speed $S = \frac{\pi dN}{1000}$ M/min

EXAMPLE 1

Find the cutting Speed of a 50mm diameter drill running at 955 rev/min of 178 rev/min

SOLUTION

$$S = \frac{\pi dN}{1000}$$

$$= \frac{\pi \times 50 \times 178}{1000}$$

$$= \underline{\underline{28\text{m/min}}}$$

EXAMPLE 2

Find the cutting Speed of a 15mm diameter drill running at 955 rev/min.

$$S = \frac{\pi dN}{1000}$$

$$= \frac{\pi \times 15 \times 955}{1000}$$

$$= \underline{\underline{45\text{m/min}}}$$

Knowing the Value of the cutting Speed for a particular Combination of cutting tool Material and work Material and ~~work~~ Knowing the diameter of the work to be produced (or the total diameter) the unknown Value is the Spindle Speed N at which the work or tool Should be run. From the equation of Cutting Speed, this is given by

$$N = \frac{1000S}{\pi d}$$

EXAMPLE 1.

At what Spindle Speed would a 200mm diameter high Speed - Steel Milling Cutter be run to Machine a Steel workpiece, if the cutting Speed is 28 m/min.

SOLUTION

$$N = \frac{1000S}{\pi d}$$

$$= \frac{1000 \times 28}{\pi \times 200}$$

$$= \underline{\underline{45 \text{ rev/min}}}$$

EXAMPLE 2.

What Spindle Speed would be required to turn a 150mm diameter Cast Iron Component using Cemented - tungsten - carbide tooling at a cutting Speed of 160m/min.

SOLUTION.

$$N = \frac{1000S}{\pi d}$$

$$= \frac{1000 \times 160}{\pi \times 150}$$

$$= 340 \text{ rev/min}$$

MILLING MACHINE.

Milling is the process of Machining flat, Curved, or irregular surfaces, by feeding the workpiece against a rotating cutter containing a number of cutting edges. The Milling Machine consist basically of a motor driven spindle, which Mounts and revolves the milling cutter, and a reciprocating adjustable worktable, which mounts and feed the workpiece.

Milling Machines are classified as Vertical or horizontal. These machines are also classified as knee type, ram type, Manufacturing or bed type and planer type. Most Milling Machines have self contained electric motor drive, coolant systems, variable spindle speeds, and power operated table feeds.

## CAPACITIES OF MILLING MACHINE

The capacity of milling machine is mainly determined by the area of the workpiece surface and the longitudinal travel of the table.

## MILLING ACCESSORIES

These includes the dividing head, Vertical milling attachment, Spiral milling attachment and slotting attachment.

## MILLING CUTTERS.

These includes plain milling cutters, side milling cutters, end mills, face milling cutters, slitting saws, angle milling cutters and form milling cutters.

## CUTTER HOLDING DEVICES.

These includes the arbor, adaptors, chucks and collars.

## WORK HOLDING DEVICES.

These includes. T-Slots on the worktable, T-Slots, Straps, Clamps, Solid blocks, Step blocks, angle plate, parallel strip and plain machine vice, universal vice, rotary table and dividing head.

## DIVIDING HEAD.

A dividing head is used to change the angular

Position of a work piece in relation to the cutter in a precise and pre-determined manner. By its use, the following milling operations can be done accurately; milling grooves on the surface of revolutions such as flutes of reamer drill and taps.

Most dividing heads have 40 teeth on the worm wheel, and the worm is single threaded such that the worm (or crank) is turned 40 times to obtain one complete revolutions of the spindle.

TYPES OF INDEXING.

There are the types of indexing; simple indexing or direct indexing, compound indexing, differential indexing and angular indexing.

SIMPLE INDEXING.

This is accomplished by means of the direct index plate located on the spindle nose. It is rapid method of indexing which is used for some types of milling operations such as milling of a square nut, milling hexagonal, fluting or grooving, milling taps and similar work requiring a small number of divisions.

However, if we require (n) number of divisions equally on a work piece, each division will require 40/n revolutions of the index crank since 40 turns of the crank cause (1) complete revolutions of work.

Therefore:

$$\text{Indexing} = 40/n = C\frac{a}{b} = C + a/b \times \frac{M}{M} = C + a/b$$

$\Rightarrow$ where $C$ = whole turn of the crank and $a$, $b$ and $M$ are integers.

EXAMPLES

1. Determine suitable indexing for the following

numbers of divisions.
(a) 4, (b) 16, (c) 33, (d) 6 and (e) 60.

Solutions

From the above equation $40/n$:

(i) 4 divisions
Indexing $= 40/4 = 10$ whole turns

(ii) 16 divisions
Indexing $= 40/16 = 2\,8/16$ 2 whole turns and 8 holes in a 16 hole circle.

(iii) 33 divisions
Indexing $= 40/33 = 1\,7/33 = 1$ whole turn and Seven (7) holes in 33 hole circle

(iv) 6 divisions
Indexing $= 40/6 = 6\,4/6 = 6\,2/3 = 6 + \frac{2\times7}{3\times7} = 6\,\frac{14}{21}$

OR

$6\,\frac{2\times8}{3\times8} = 6\,\frac{16}{24} =$ ~~6 holes in a 24~~
6 whole turn and 16 holes in a 24 hole circle.

(v) 60 divisions
Indexing $40/60 = 2/3 = 0 + \frac{2\times6}{3\times6} = 0 + \frac{12}{18}$ or $0 + \frac{14}{21}$
$= 0 + \frac{16}{24}$

∴ This result = 0 whole turns and 14 holes in a 18 hole circle etc.

MILLING CUTTERS

These includes plain Milling Cutters, Side Milling Cutters, end Mills, Face Milling Cutters, Slitting Saw, angle Milling Cutters and form Milling Cutters.

FIRE.

Fire is a phenomenon in which Combustible Material especially organic Materials containing Carbon, reacts chemically with oxygen in the air to produce heat. Flame arises from the combustion of volatile liquids and gases evolved and spreads the fire. NO one should under estimate the danger of fire. Many Materials burns rapidly and the fumes and smoke produced, particularly from Synthetic Materials, including plastics May be deadly.

There are number of reasons for fire Starting

(i) Malicious Ignition: i.e deliberate fire raising, misuse of faulty electrical equipment, e.g. incorrect plugs and wiring, damaged cables, over loaded Sockets and Cables, Sparking and equipment such as soldering irons left on and un attended.

(ii) Cigarette and Matches: Smoking in unauthorised areas, throwing away lighted Cigarettes or Matches.

(iii). Mechanical heat and Sparks: eg faulty Motors, overheated bearings, spark produced by grinding and cutting operations.

(iv) Heating plant: flamable liquid/substance in Contact with hot Surfaces.

(v) Rubbish burning: Casual burning of waste and rubish.

FIRE PREVENTION.

The best fire prevention is ~~St~~ to Stop a Starting of fire. However, fire can be prevented by the following ways.

(i) Where possible use Materials which are less flammeable.

(ii) Minimize the quantity of flammeable Materials kept in the workplace or Store.

(iii) Store flammeable Materials Safely, well away from hazardous processes or Materials, and ~~build~~ where appropriate from buildings.

(iv) Warn people of the fire risk by a conspicous Sign at ~~which~~ each work place, Storage area and on each Container.

(v) Some items, like oil Sealed rags May ignite Spontaneously. Keep them in a Metal Container away from other flammeable Material.

(vi) before Welding or Similar work remove or insulate flammeable Materials and have fire extinguishers to hand.

(vii) Control ignition Sources, eg nake flames and Sparks and Make Sure that no Smoking rules are obeyed.

These and Many more are the Sources of fire prevention method.

FIRE FIGHTING

1. Every employee Should know where the portable fire extinguisher, the hose reels and the Controls for extinguishers in the working area. This training Must include the use of extinguisher or Simulated fires.

2. It Must be Stressed that the fire fighting Should only be attempted if it is Safe to do So and that an escape route Must always be available.

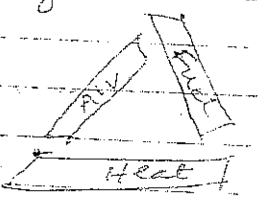
(10)

(3) It is also essential to emphasis the limits of first aid fighting in order to show the need to attempt this safely and the importance of first araising the alarm.

(4) As previously stated, fire requires fuel, oxygen and heat. This is shown by the fire triangle in the figure below, where one side stands for fuel another for heat and the third for air or oxygen. If any one side is removed the fire inside will go out.

(5) The extinguishing of a fire is generally brought about by depriving the ~~oxygen~~ burning substance of oxygen.

(6) By far the most important extinguishing agent, by reason of its availability and generally effectiveness is water. It is more effective than any other common substance in absorbing heat and thereby reducing the temperature of the burning mass.



Triangle of fire

CLASSES OF FIRE

(i) CLASS A: This is ordinary combustible fire, they are materials that are used as a fuel, e.g. solid materials such as wood, paper, fibres, trash etc.

(ii) CLASS B: This is the fire that uses flammeable liquid as the source. This includes petroleum bad oils, kerosene and gasoline. Fable gases such as butane or propane are also common fuel sources.

(iii) CLASS C: This is fire that uses electrical Components, eg electric Motors, Generators, Switches panel or boxes and electrical trans-former.

(iv) CLASS D: This is the fire that uses Metals as Combustible Materials, eg titanium, Magnesium Aluminium and potassium.

(v) CLASS K: This is the fire that results from (Cooking fire), eg Cooking oils, Vegetable fats, animal fats etc.

~~METAL CUTTING~~

## SAFETY

This is defined as all Measures taken to ensure the general well being of the worker and prevent Machine down time and Material wastage due to accidental damages in order to raise the profitability of the enterprise.

## OBJECTIVES OF HEALTH AND SAFETY ACT 1974.

(i) To Secure the health, Safety and Welfare of people at work.

(ii) To protect other people against risk to health or Safety arising from the activity of people at work.

(iii) To control the Keeping and use of dangerous Substances and prevent people from un-lawfully having or using them.

(iv) To control the release into the atmosphere of noxious or offensive Substances from prescribed premises.

## EMPLOYEE'S RESPONSIBILITY.

(i) To take reasonable care for their own health and Safety and that of others and sprays), dust, gases and welding arcs. Eye protectors includes Safety Spectacles, eye Shields, goggles, Welding filters, face Shields and hoods.

HEAD PROTECTION.

This includes industrial Safety helments to protect against falling objects or impact with fixed objects, industrial Scalp protectors to protect against Striking fixed obstacles, Scalping or entanglement, and Caps and hairnets to protect against Scalping and entanglement.

FOOT PROTECTION.

This includes Safety boots or Shoes with Steel toe caps, foundry boots with Steel toe caps which are heat resistance and designed to keep out Molten Metal. Wellington boots to protect against Water and wet conditions and antistatic foot wear to prevent the built up of Static electricity on the Wearer.

PROTECTIVE CLOTHING.

This is the type of clothing used for body protection and this includes Cover ralls, overalls and aprons to protect against chemicals and other hazardous Substances. Out fit to protect against cold, heat and bad Weather and clothing to protect against Machinery Such as chainSaws.

GOOD HANDLING TECHNIQUES.

(i) STOP AND THINK: plan the lift. organise the work to minimise the amount of lifting necessary. Know where you going to place the load. Use Mechanical assistance if possible. Get help if load is too heavy. Make sure your path is clear. Don't let the load obstract your view. For a long lift, i.e from floor to shoulder height, consider a rest midway on a bench in order to adjust your grip. Alternatively lift from floor to knee then from the knee to carrying position. Reverse this method when setting the load down.

(ii) PLACE YOUR FEET: Keep your feet apart to give a balance and stable base for lifting. Your leading leg should be as far forward as comfortable.

(iii) ADOPT A GOOD POSTURE: Keep your back straight and upright. Bend your knees and let your legs do the work. Keep your shoulders level and facing the same direction as your hips. Don't twist your body.

(iv) GET A FIRM GRIP: Lean forward a little over the load to get a good grip. Try to keep your arms within the boundry formed by your legs. balance the load using both hands. A hook grip is less tiring than keeping the fingers straight. Wear gloves if surface is rough or has sharp edges. Take great care is the load is wrapped or slippery in anyway.

(v) DON'T JERK: Carry out lifting operations smoothly, keeping control of the load. Move the feet, don't twist your ~~feet~~ body if you turn to the side.

(vi) KEEP CLOSE TO THE LOAD: If the load is not close when lifting, try sliding it towards you. Keep the ~~load~~ heaviest side of the load who may be affected by what they do or don't do.

This duty implies not only avoiding Sily or reckless behaviours but also understanding hazards and complying with Safety rules and procedures. This means that you correctly use all work item provided by your employer in accordance with the training and instruction you received to enable you to use them safely.

(ii) To cooperate with their employer on health and Safety.

This duty means that you should inform, without delay of any work situation which might be dangerous and notify any short comings in health and Safety arrangements so that remedial action may be taken.

EMPLOYERS RESPONSIBILITY.

(i) To provide and maintain machinery, equipment and other plant, and systems of work that are Safe and without risk to health. Systems of work means that, the way in which the work is organised and includes layout of the workshop, the other in which jobs are carried out, or special precaution to be taken before carrying out certain hazardous tasks.

(ii) Ensure ways in which particular articles and substances eg machinery and chemicals are used, handled, stored and transported are Safe and without risk to health.

(iii) Provide information, instruction, training and supervision necessary to ensure health and Safety at work. Information Means the background knowledge needed to put the instruction and training into context. Instruction is when Some one show others how to do Something by practical demonstration. Training Means having employees practice a task to improve their performance. Suppervision is needed to oversee and guide in all Matters related to the task.

(iv) Ensure any place under their Control and Where their employee work is Kept in a Safe Condition and doesn't pose a risk to health. This includes into and out of the work place.

(v) Ensure the health and Safety of their employees Working environment eg heating, lightening, Ventilation are provided. They must also provide adequate arrangement for the Welfare at work of their employees. The term of Welfare at work covers facilities Such as Seating, Washing, toilets etc.

## PERSONNEL PROTECTIVE EQUIPMENT(PPE)

This is defined as all equipment which is intended to be worn or held to protect against risk to health and Safety. These include Most types of protective clothing and equipment Such as eye, head, foot and Hand protection and protective clothing for the body.

## EYE PROTECTION

This Serves as guard against the hazard of impacts, Splashes from chemicals or Molten Metals, liquids droplets (chemical Mist and

## MEASUREMENT

Measurement is the process of experimentally obtaining one or more values of a physical quantity that can reasonably be attributed to the quantity. Measurement generally involves the use of measurement (measuring) instrument (measurement system) as a means of determining an unknown quantity or variable called the measurand.

In another words, the measurement of a given physical quantity or variable is the result of comparison between the physical quantity whose magnitude is unknown (called the measurand) with a similar quantity whose magnitude is known called the standard.

The standard is the physical embodiment of the unit of measurement and its value. The standard of comparison must be of the same character or type as the measurand. It is usually prescribed and defined by a legal or recognized agency or organization such as ISO, ANSI, BS etc
(International organization for Standardization, American National Standards Institute, British Standard)

# Testing.

Testing Can be defined as Subjecting of Something in Other to determine how well it works.

Testing Can Simply be defined also as the act of Examination of the nature or the Value of Something. A testing is Carried out to ensure that a product Conforms to its specification. It is also the method used in making such an Examination. Testing is used in various aspects in the aspect of Idealing testing method.

**Instrumentation:** Is a Collective term for measuring instruments used for indicating, measuring and recording physical quantities.

**The need for Measurement.**

1. To ensure that the part to be measured conforms to the established standard.
2. To meet the interchangeability of manufacture
3. To provide customer satisfaction by ensuring that no faulty product reaches the customers.
4. To coordinate the functions of quality control, production, procurement and other department of an organization
5. To judge the possibility of making some of the defective parts acceptable after minor repairs.

. Measurement play a very significant role in every branch of scientific and engineering researches and processes e.g in control systems, process instrumentation etc.

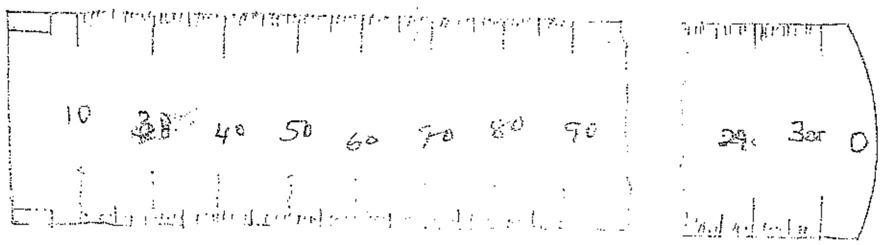
## LINEAR MEASURING INSTRUMENTS:

1. Scale (straight edge graduated rule, tape etc)
2. Caliper.
3. Vernier Caliper (Vernier height gauge and Vernier depth gauge)
4. Micrometer
5. Comparators (Dial gauges or Dial indicators.
6. Interferometer (optical flats)

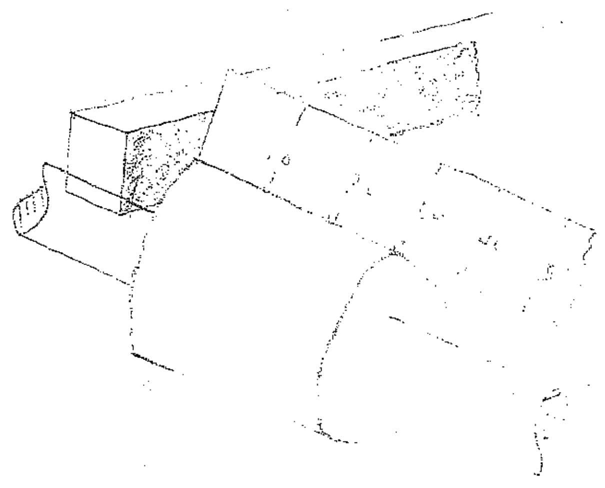
## Simple Measuring Exercises Using,

### 1. Steel Rule (Engineer's Rule)

The steel rule, as the name implies, is made of stainless steel. it normally has two metric scales each with accuracies of 1mm and 0.5mm. They come in various lengths such as 150mm, 300mm, 500mm or 1000mm.




a) Steel rule of 300MM



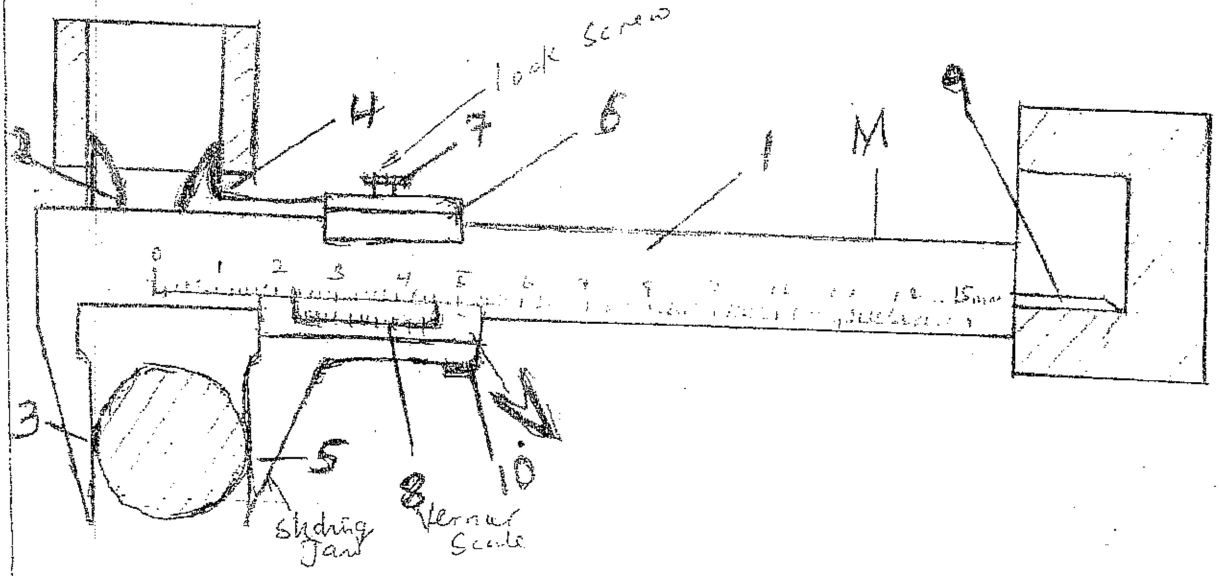
b) Butting the end of the rule (steel rule).

### 2. Vernier Caliper

The Vernier is a beam caliper with jaws at right angles to the beam and is used for measuring external and internal dimensions and the depth of slots and holes.

12

The beam (1) carries (2) measuring jaws (2) and (3) while the other two jaw (4) and (5) are made integral with head (6) with which they slide freely along the beam. Head (6) is clamped in place by screw (7). Head (6) carries plate (8) with the Vernier Scale graduations. The narrow depth attachment (9) is used to measure depths. It is rigidly attached to head (6) and is moved in the groove of the beam along which the head (6) is shifted by the knurled button (10).



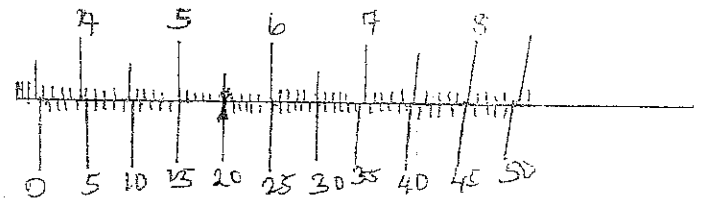

THE VERNIER CALIPER

The Vernier Caliper shows three applications, Jaws (3) and (5) measure the diameter of the shaft, Jaws (2) and (4) measure the diameter of a hole and attachment (9) is used to measure the depth of a slot.

(5)

Example

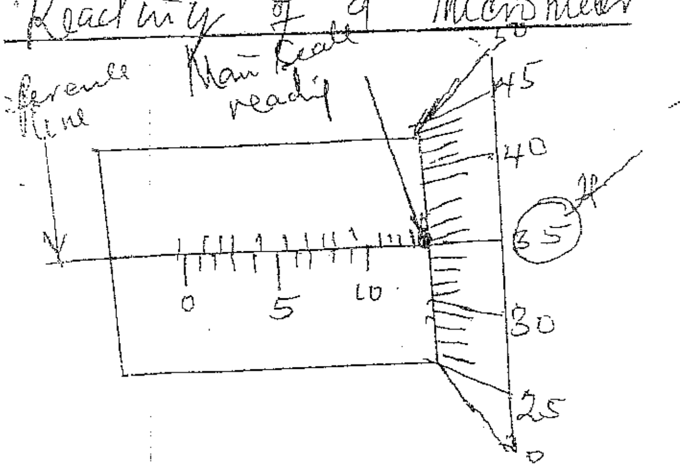
find the reading of the Measurement Shown by Vernier Caliper below.




Least Count: $\frac{1}{50} = \underline{0.02}$

- from the above the Main Scale reading is 35mm
- The Vernier division Coinciding with the Main Scale is the 20th division
- Multiply this number (20) with the Vernier Constant or least Count (L.C = 0.02mm) i.e $20 \times 0.02mm = 0.40mm$, This will give the Vernier Scale reading.
- The total reading will be the Main Scale reading (35) plus the Vernier Scale reading (0.40mm) i.e $35mm + 0.40mm = 35.40mm$.

## Reading of a Micrometer




Draw a diagram of micrometer screw gauge and label its

The smallest division on the thimble is 0.5, and the no of division on the thimble 50.

The least Count of micrometer is $\frac{0.5}{50} = 0.01mm$

- The number of Main Scale division in millimeters above the reference line or datum line on the barrel is noted i.e 13.5mm
- The number of Main Scale sub-division below the reference line on the barrel exceeding only the upper graduation is also noted.
- Adding the number of division above the reference line and the number of division below the reference line will give the Main Scale reading.
- find the number of division on the thimble and multiply it with least Count (L.C) of the micrometer.
- Adding the Main Scale reading with the product above (number of divisions on the thimble multiply by the least Count (L.C) of the micrometer) will give

the reading of the micrometer.

Total reading = Main Scale reading + (LC × reading of thimble)

= 13.5 + 0.01 × 35

Reading of a micrometer = 13.85 mm

So, from the above, the micrometer shows a reading of 13.85 mm, it means there are:

- 13 divisions above the reference line = 13.00 mm
- 1 division below the reference line = 0.50 mm
- 35 thimble divisions = 0.35 mm
- The reading of the micrometer is 13.85 mm.

i.e 13.00 + 0.50 + 0.35 = $\underline{\underline{13.85}}$ m

## METAL JOINING PROCESS

Some methods of joining parts together is used throughout industry to form either a complete product or an assembly. The method used depends on the application of the finished product and whether the parts have to be dismantled for maintenance or replacement during service.

There are five methods by which part may be joined, these are:

1. Adhesive bonding.
2. Soldering
3. Brazing
4. Welding
5. Mechanical fasteners, using screws, bolts and nut, rivets.

### 1. ADHESIVE BONDING.

Adhesive are used to establish a permanent bond between two joints surfaces, Adhesive bonding is the modern term for glueing. Many types of adhesive are available, there applications varying from use in surgery to heal internal wounds and organs to joining structure members in aircraft.

SOLDERING.

Soldering is a method of joining metals by using an alloy having a low melting point than the metal being joined and which unites fully with them. Soldering is always refers to as soft-soldering with melting temperature less than 427°C. Soldering iron and solder is used in the process of soldering.

3. BRAZING.

Brazing is a welding process where in coalescence (come together and unite to one another) is produced by heating to suitable temperature and by using a filler metal having the solidus of above 427°C and below the solidus of the base metals.

4. WELDING.

Welding can be defined as a localized coalescence of metals where in coalescence is produced by heating to suitable temperature, with or without the application of pressure and with or without the use of filler metals.

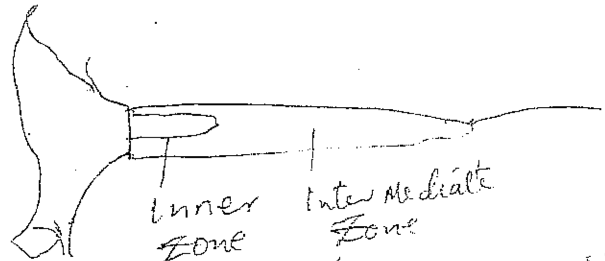
5. MECHANICAL FASTENERS

Mechanical fasteners are used to establish temporary or permanent joint, by using bolts and nut, rivet etc.

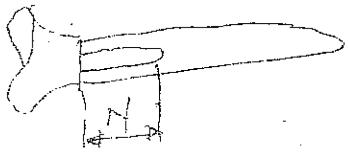
# WELDING

Type of flames in Oxy-acetylene:

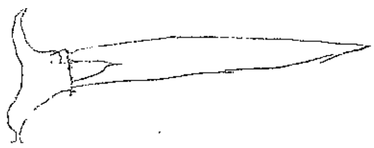
Flame Structure



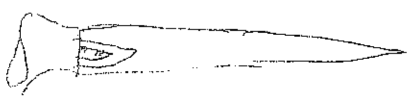

1. Neutral flame




2. Oxidising flame



3. Carburizing flame



[illegible] surface. The plunge is perpendicular to the Measuring

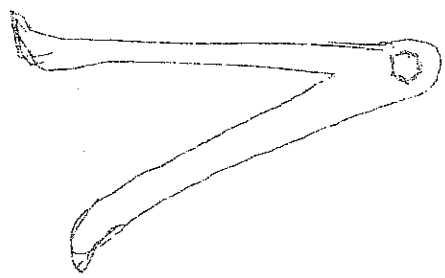
(17)

## Of Calipers

There are two types of Calipers. One is called inside Calipers and the other one is called outside Calipers.

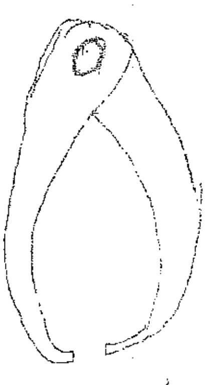
) Inside Calipers

~~the~~ Inside Calipers are used to measure the inside diameter of a hole or in general to measure the inside dimensions of any object within the capacity of this tool.



(B) Outside Calipers

Outside Calipers are used to measure the outside diameter of a round bar or in general to measure the outside dimensions of any object within the capacity of this tool.



outside Calipers

# FLAME TYPES EQUATIONS (43) (2)

Three types of flames may be obtained from the oxy-acetylene flame, namely Neutral, oxidizing and Carburizing flame.

The neutral flame or balance flame is obtained by burning one part of acetylene to one part of oxygen by volume to 2Co and $H_2$ as given in equation.

$C_2H_2 + O_2 = 2Co + H_2 + 449.2$ KJ/g-mole.

The oxidizing flame is obtained by burning one part acetylene with excess or greater than one part of oxygen.

$C_2H_2 + \cancel{1}.5O_2 = 2Co + H_2 + 0.5O_2$.

The Carburizing flame is obtained by burning excess acetylene, say one part of acetylene to less than one part oxygen.

$C_2H_2 + 0.5O_2 = Co + C + H_2$.

## Fluxes

A thin film of oxide is always formed, especially at elevated temperatures, on metal surface - which if carried over into the weld metal will prevent a reliable union between the parent metal and filler metal. As a result of this, a flux is used to dissolve the oxide. The flux also serves to release trapped gases and slag from the weld. We employ a flux which should fulfil the following conditions:

i) it should easily melt and have a lower melting temperature than ~~the~~ the parent and filler metals.

ii) it should readily react with metallic oxides so that oxides are completely dissolved by the time the molten pool solidifies.

iii) its specific gravity should be lower than that of the parent and filler metals so that the slag formed would easily float to the surface of the weld pool. The slag must reliably protect the weld metal from atmospheric oxidation and come off easily after solidification.

(iv) it should in the molten state readily spread over the weld pool.

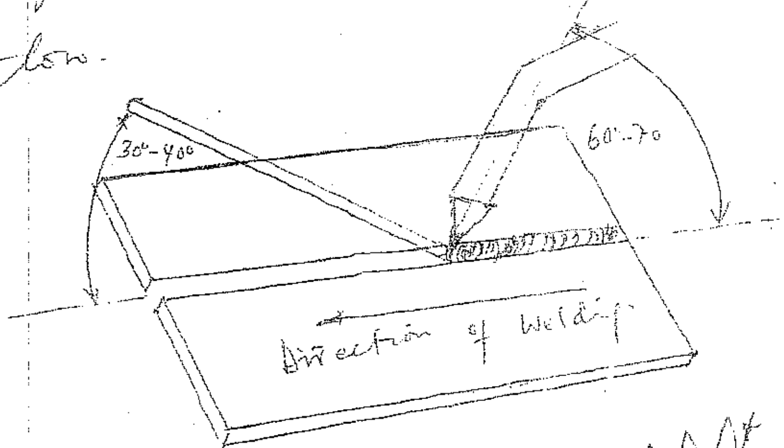
v) To remain stable (not to deteriorate) even if the heating is prolonged.

# GAS WELDING TECHNIQUE (44) ~~(4)~~ (47)

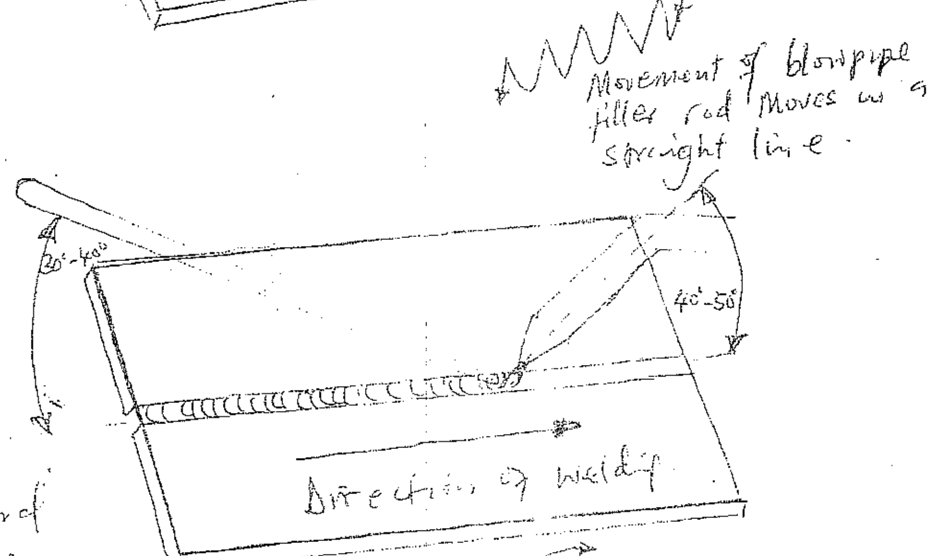
Two techniques of torch movement are used in gas welding. These are the leftward welding and rightward welding techniques.

In the leftward technique, the welding rod precedes the torch along the seam and the weld travel from right to left as shown below.

In the rightward welding technique the weld progresses along the seam from left to right, the filler rod following the torch as shown below.




Leftward welding.




Rightward welding.

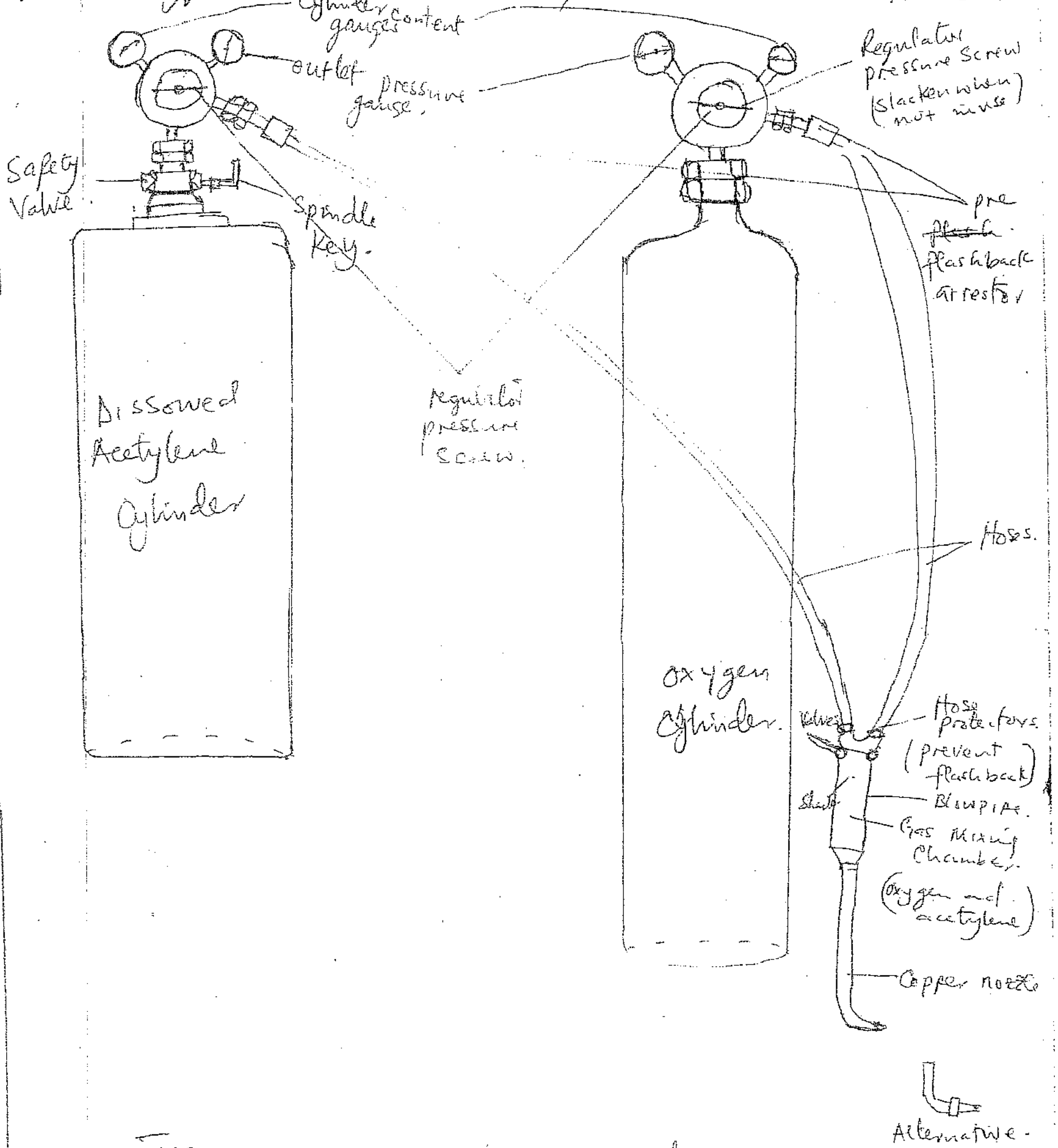
TECHNIQUES OF GAS WELDING.

## THE WELDING TORCH OR BLOW-PIPE. (5)

There are two types of welding blowpipes, high pressure type and low pressure type. The high pressure blowpipe is used with high pressure equipment in which the acetylene gas is supplied from cylinders. it consists of a shank, control valves, mixing chamber and nozzle or tip. The high pressure blowpipe is just a mixing device to supply approximately equal volumes of oxygen and acetylene to the nozzle. the control valves are used to vary the pressure of the gases as required. The nozzle supplies the flame at the end of the blow pipe; and there are two types: detachable tip or nozzle and swaged nozzle.

Both types perform the same function, how ever, the swaged type is slightly cheaper and it's easier to fit. Nozzle comes in various sizes determined by the diameter of the hole drilled in the end of the nozzle. The nozzle orifice diameter determines the operating gas pressure with which to obtain

the best flame conditions. ~~(b)~~ (b)
. The correct nozzle size in use is determined by the type and thickness of metal to be welded.



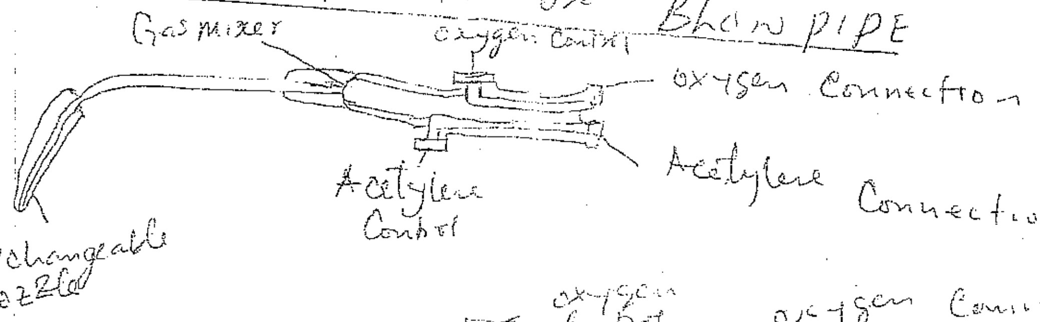

THE OXY-ACETYLENE WELDING SET.

## TYPES OF FLAME USED FOR WELDING VARIOUS METALS

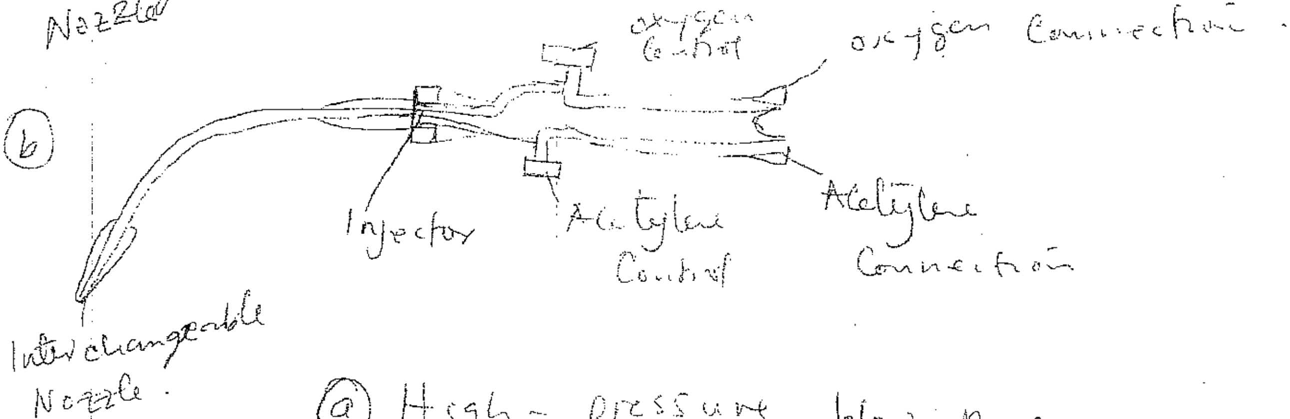
| METALS | FLAME |
|---|---|
| Low Carbon Steel | Neutral |
| Stainless Steel | Neutral |
| Cast Iron | Neutral |
| Copper | Neutral |
| Aluminum | Neutral |
| Nickel alloys | Neutral |
| Monel | Neutral |
| Everdur | Neutral |
| Brass | oxidizing |
| Inconel | Slightly Carburizing |
| Aluminium alloys | Slightly ,, |
| Magnesium alloys | Slightly ,, |

## WELDING TORCH OR BLOW PIPE

(a)




(b)




(a) High - pressure blow pipe.

(b) Low - pressure blow pipe.

(6)

11

## ELECTRIC ARC WELDING (5)

Electric arc welding processes depend on the existence of an electric arc, which supplies the heat necessary for the localised melting of material at the joint of parent metals. An electric arc is obtained when an electric current flows between two electrodes separated by a short distance. In electric arc welding one electrode is the welding rod and the other is the workpiece being welded.

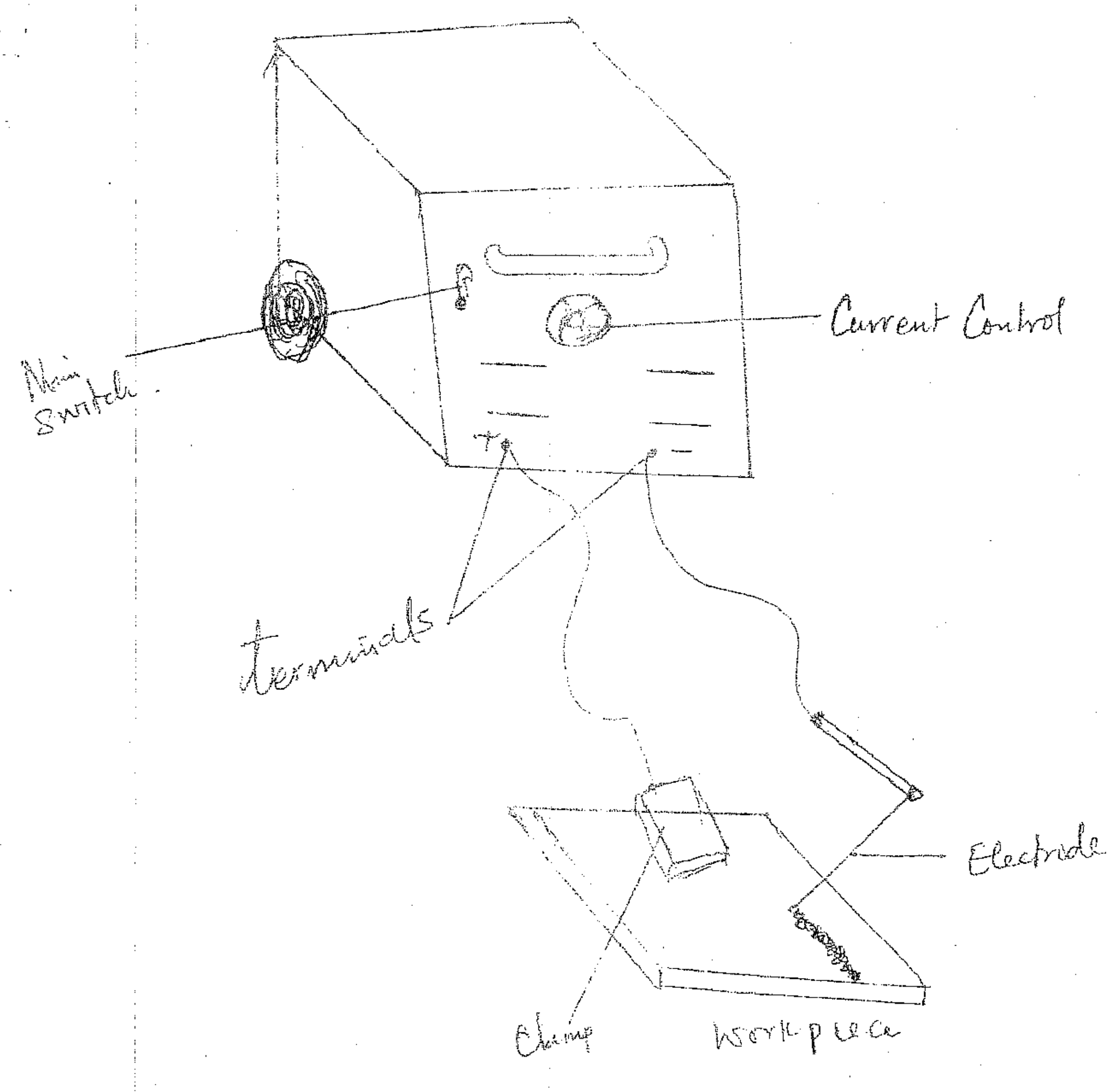
Arc Welding Machines

Arc welding machines are of many types due to the many arc welding processes available. Selection of the welding m/c depends largely on the type of application.

The arc welding machine is usually a transformer which converts for example the power supply of 230 volts and low power current of 10 to 100 amperes to that suitable for arc welding for a motor / engine generator.

The type of welding voltage used by a welding m/c may be alternating current (a.c) or direct current (d.c).

(8)

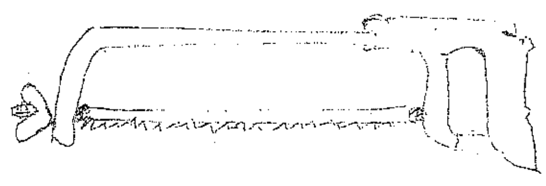



A WELDING TRANSFORMER

## HAND, BENCH, AND MACHINES CUTTING TOOLS

(16)

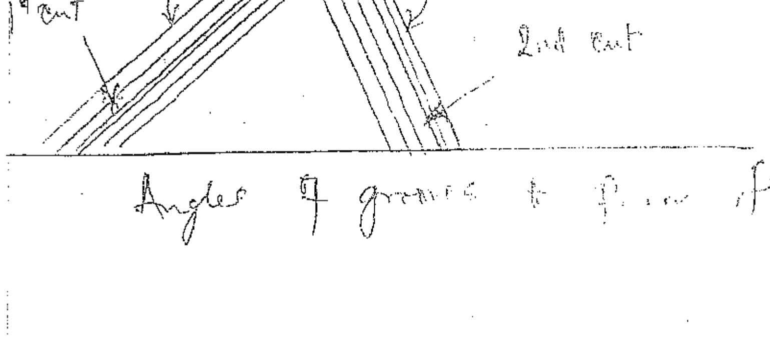
### The Hacksaw



The hacksaw is the chief tool used by the fitter for cutting off Metal, There are Many sizes and types of hacksaw blades on the market with special recommendations for particular classes of work and in view of the variable factors governing the selection of a blade it is advisable to keep a stock of different blades to meet the different Sawing Conditions.

### File

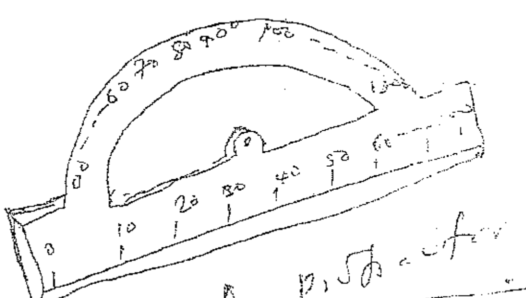
The most usual form of file is that with cross cut teeth, the grooves in the face of the file running in two direction and dividing it up into small diamond-shaped teeth.




Angles of grooves to form file teeth.

Q. Protractors    LP 114

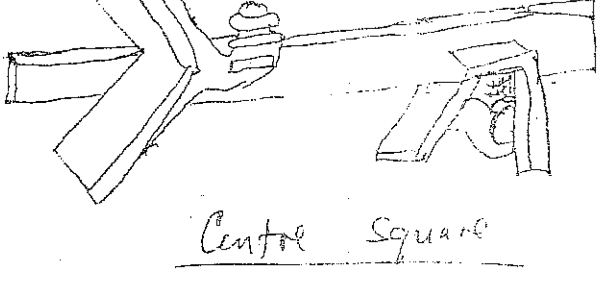
An instrument for measuring angles in metal work. The try square can be used for checking a 90° or right angle only but the protractor can be used for checking any angle.




A protractor

Q. Centre Square

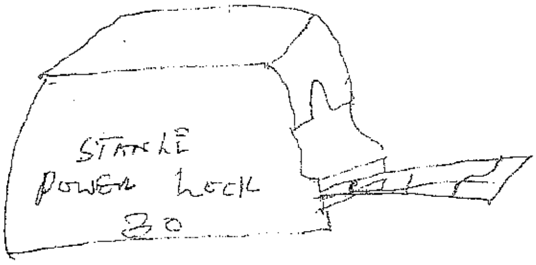
Used for drawing or marking lines across the end of a round bar. When two lines are drawn across the end of a bar at a 90° angle to each other with this tool, the point where one line crosses the other is the centre of the round bar.



Centre Square

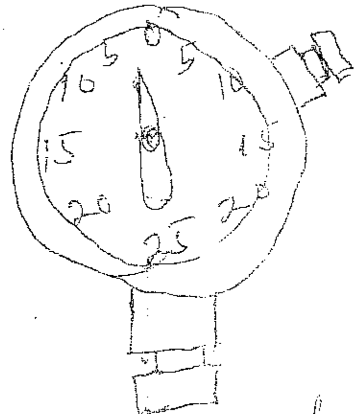
⑤ Tape rule

it is used for measuring lenghts of over 100m




⑥ Dial indicator-

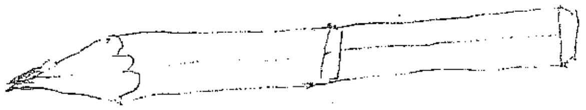
it is used to measure small distance and angles and amplify them. it is used to check the variation in tolerance during the inspection process of a machined part. It typically measures ranges from 0.25mm to 300mm (0.015 in to 12.01 in) with graduation of 0.001mm to 0.01mm (metric) or 0.00005 in to 0.001 in imperial.




Dial indicator are used to check flatness, straightness and concentricity. The indicator is mounted on a stand

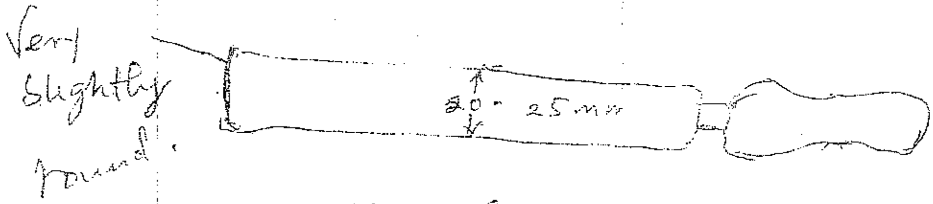
## 3. The Centre punch (18)

The Centre punch is used when circular dot marks are required.

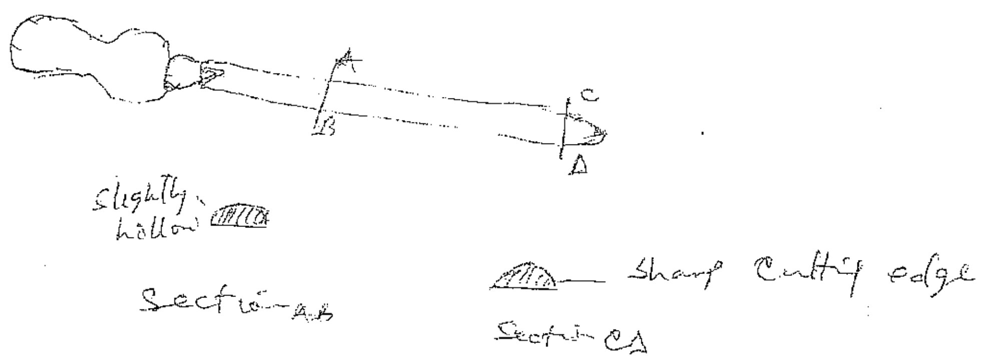


## 4. The Scrapers and Scraping

We have two main types of scrapers
They are a) flat Scraper b) Half-round scraper.




Flat Scraper.




The Scraper is a tool used to make the Surface or face of workpiece Smooth finish or for the purpose of removing errors in flatness.

Assignment 2; (19)

1) with the aid of diagram draw all Measuring Instruments.

FEELER GAUGE

feeler gauge are used for checking small clearances and can also be used for measuring clearances between, contact breaker, plug gaps, valves setting, etc

Marking out tools (20) and operations

1. Surface plate
2. scriber
3. Divider
4. odd-leg caliper (Hermaphrodite)
5. Try square
6. Straight edge
7. Vee block
8. Surface gauge
9. Angle plate